

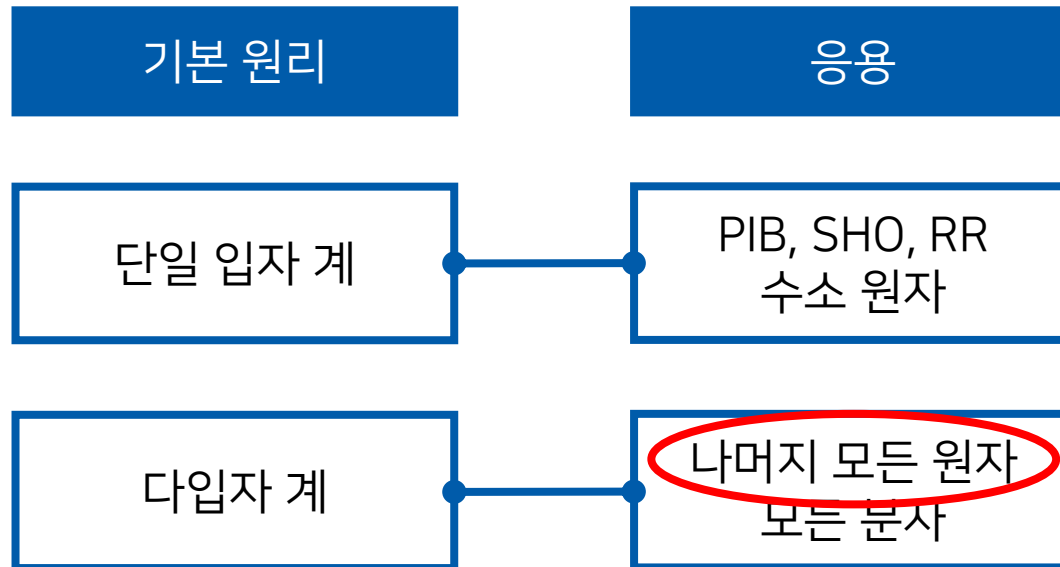
Physical Chemistry (II)

Lecture 17

Hartree-Fock Method

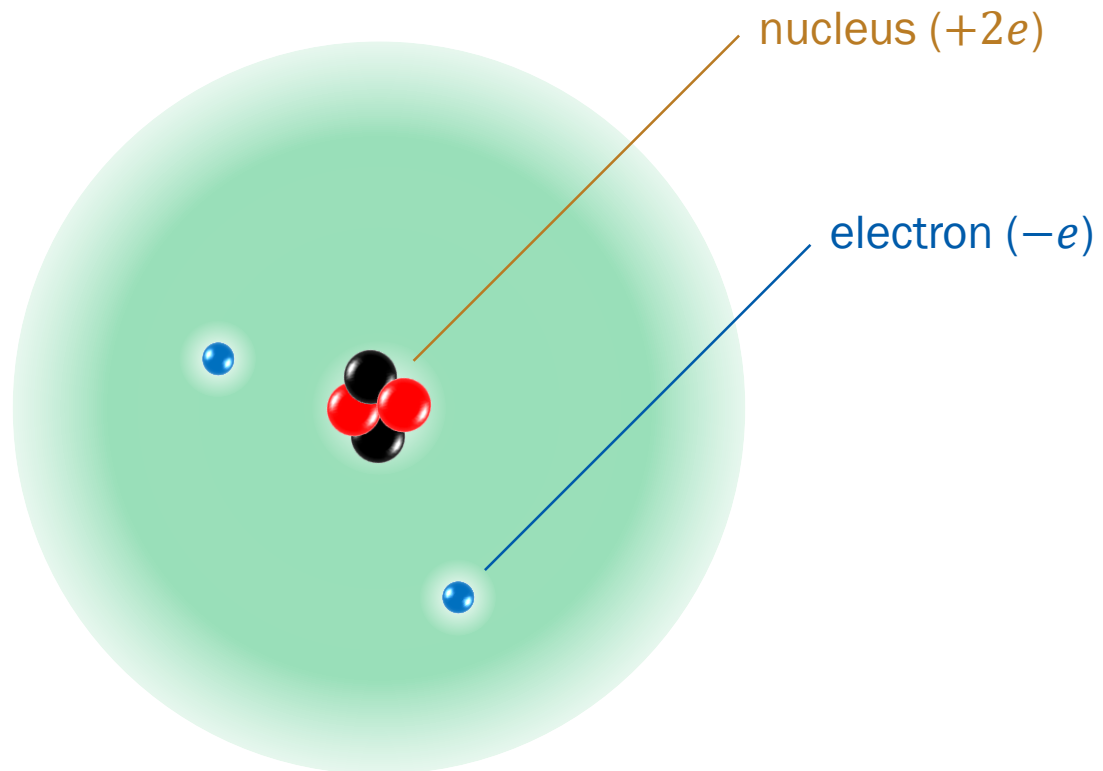


Where Are We?



RS 7.1
M 7.9

Helium Atom



Ground-State Energy of He

Method	Energy (E_h)	Method	Energy (E_h)
<i>Complete neglect of the interelectronic repulsion term</i>			-4.0000
<i>Perturbation calculations</i>		<i>Variational calculations*</i>	
First-order	-2.7500	1s orbitals (1)	-2.8477
Second-order	-2.9077	Eckart (2)	-2.8757
13 th -order	-2.90372433	Hartree-Fock (2)	-2.86168
		Hylleraas (10)	-2.90363
		Pekeris (1078)	-2.903724375
<i>Experimental value</i>			-2.9033

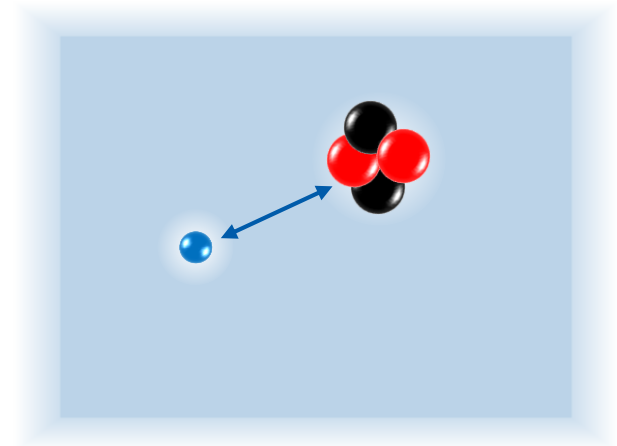
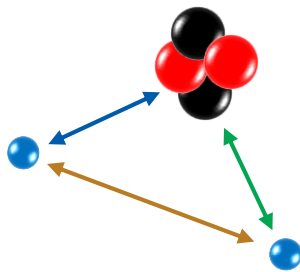
* the number in parentheses indicates the number of parameters

Treating Helium

- Perturbation and variational approaches
 - can give very accurate energies
 - computationally difficult
 - not transferable to large atoms and molecules
- Orbital approximation
 - Each electron's spatial part is represented by an one-electron orbital $\phi_i(\vec{r})$
 - Common concept in chemistry
 - How to obtain $\phi_i(\vec{r})$?

Mean-Field Approach

$$\hat{H} = \underbrace{-\frac{\hbar^2}{2m_e} \nabla_{e1}^2 - \frac{2e^2}{4\pi\epsilon_0 r_1}}_{\hat{H}_{H1}} - \underbrace{\frac{\hbar^2}{2m_e} \nabla_{e2}^2 - \frac{2e^2}{4\pi\epsilon_0 r_2}}_{\hat{H}_{H2}} + \underbrace{\frac{e^2}{4\pi\epsilon_0 |\vec{r}_1 - \vec{r}_2|}}_{\text{interelectronic repulsion}}$$



Self-Consistent Field Method

- Trial function: $\phi(\vec{r}_1, \vec{r}_2) = \psi(\vec{r}_1)\psi'(\vec{r}_2)$
- Probability distribution of electron 2: $|\psi'(\vec{r}_2)|^2 d\vec{r}_2$
- Potential energy of electron 1 due to electron 2:

$$V_1^{\text{eff}}(\vec{r}_1) = \frac{e^2}{4\pi\epsilon_0} \int \frac{1}{|\vec{r}_1 - \vec{r}_2|} |\psi'(\vec{r}_2)|^2 d\vec{r}_2$$

Self-Consistent Field Method

- Hamiltonian for electron 1:

$$\hat{H}_1^{\text{eff}} = \hat{H}_{\text{H1}} + V_1^{\text{eff}}(\vec{r}_1) = -\frac{\hbar^2}{2m_e} \nabla_{e1}^2 - \frac{2e^2}{4\pi\epsilon_0 r_1} + V_1^{\text{eff}}(\vec{r}_1)$$

- Hartree-Fock equation:

$$\hat{H}_1^{\text{eff}} \psi(\vec{r}_1) = \epsilon \psi(\vec{r}_1)$$

- Now solve the HF equation to obtain updated $\psi(\vec{r}_1)$, with which one can repeat the whole procedure

One-Electron Wavefunctions

$$\hat{H}^{\text{eff}}\psi_i(\vec{r}) = \varepsilon_i\psi_i(\vec{r})$$

- Iteration gives a convergent solution: $\psi_i(\vec{r})$ and ε_i
- Still quantized (corresponding to H atom orbitals)
- Ground-state solution: $\psi_0(\vec{r})$ and ε_0
- The final two-electron solution:

$$\phi_{ij}(\vec{r}_1, \vec{r}_2) = \psi_i(\vec{r}_1)\psi_j(\vec{r}_2)$$

Example Iteration (He Atom)

- Trial function: $\phi(\vec{r}_1, \vec{r}_2) = \psi(\vec{r}_1)\psi(\vec{r}_2)$

$$\psi(\vec{r}) = c_1 \left(\frac{\zeta_1^3}{\pi} \right)^{1/2} e^{-\zeta_1 r} + c_2 \left(\frac{\zeta_2^3}{\pi} \right)^{1/2} e^{-\zeta_2 r}$$

with fixed $\zeta_1 = 1.45363$ and $\zeta_2 = 2.91093$

Iteration	c_1	c_2	ε	E
1	0.500	0.500	-0.800	-2.394
2	0.884	0.135	-0.935	-2.857
3	0.836	0.190	-0.915	-2.862
4	0.845	0.179	-0.919	-2.862
5	0.844	0.181	-0.918	-2.862
⋮				
10	0.844	0.181	-0.918	-2.862

Electronic Wavefunction

- Must contain **spin wavefunction**

$$\phi_{ij}(\vec{r}_1, \vec{r}_2; \sigma_1, \sigma_2) = \psi_i(\vec{r}_1)\psi_j(\vec{r}_2)|\sigma_1\rangle_1|\sigma_2\rangle_2$$

- Must be **antisymmetric**

$$\phi_{ij}(\vec{r}_1, \vec{r}_2; \sigma_1, \sigma_2) = -\phi_{ij}(\vec{r}_2, \vec{r}_1; \sigma_2, \sigma_1)$$

- **Pauli exclusion principle:** two identical fermions cannot be detected at the same position

Ground-State Wavefunction

$$\phi_{ij}(\vec{r}_1, \vec{r}_2; \sigma_1, \sigma_2) = \psi_i(\vec{r}_1)\psi_j(\vec{r}_2)|\sigma_1\rangle_1|\sigma_2\rangle_2$$

- The system energy only depends on the **spatial part** (unless a magnetic field is applied)
- The lowest energy is obtained when

$$\psi_i(\vec{r}_1) = \psi_0(\vec{r}_1)$$

$$\psi_j(\vec{r}_2) = \psi_0(\vec{r}_2)$$

Ground-State Wavefunction

$$\phi_{00}(\vec{r}_1, \vec{r}_2; \sigma_1, \sigma_2) = \psi_0(\vec{r}_1)\psi_0(\vec{r}_2)|\sigma_1\rangle_1|\sigma_2\rangle_2$$

- The spatial wavefunction is **symmetric**
- The spin wavefunction must be **antisymmetric**: singlet

$$\frac{1}{\sqrt{2}}(|\alpha\rangle_1|\beta\rangle_2 - |\beta\rangle_1|\alpha\rangle_2)$$

- The total wavefunction

$$\phi_{00}(\vec{r}_1, \vec{r}_2) = \frac{1}{\sqrt{2}}\psi_0(\vec{r}_1)\psi_0(\vec{r}_2)(|\alpha\rangle_1|\beta\rangle_2 - |\beta\rangle_1|\alpha\rangle_2)$$

Ground-State Energy

$$\hat{H}_0 = \hat{H}_{H1} + \hat{H}_{H2}, \quad \hat{H}' = \frac{e^2}{4\pi\epsilon_0|\vec{r}_1 - \vec{r}_2|}$$

- Zeroth-order energy

$$\hat{H}_0\psi_0(\vec{r}_1)\psi_0(\vec{r}_2) = (E_0 + E_0)\psi_0(\vec{r}_1)\psi_0(\vec{r}_2)$$

- First-order energy

$$E^{(1)} = \int \psi_0^*(\vec{r}_1)\psi_0^*(\vec{r}_2)\hat{H}'\psi_0(\vec{r}_1)\psi_0(\vec{r}_2) d\vec{r}_1 d\vec{r}_2 = J_{00}$$

$$\rightarrow E = 2E_0 + J_{00}$$

Orbital Energies

$$\hat{H}_1^{\text{eff}} \psi_i(\vec{r}) = \varepsilon_i \psi_i(\vec{r})$$

$$\hat{H}_1^{\text{eff}} = \hat{H}_{\text{H1}} + V_1^{\text{eff}}(\vec{r}_1)$$

- Zeroth-order energy

$$\hat{H}_{\text{H1}} \psi_0(\vec{r}_1) = E_0 \psi_0(\vec{r}_1)$$

- First-order energy

$$E^{(1)} = \int \psi_0^*(\vec{r}_1) V_1^{\text{eff}}(\vec{r}_1) \psi_0(\vec{r}_1) d\vec{r}_1$$

First-Order Energy

$$E^{(1)} = \int \psi_0^*(\vec{r}_1) V_1^{\text{eff}}(\vec{r}_1) \psi_0(\vec{r}_1) d\vec{r}_1$$

$$V_1^{\text{eff}}(\vec{r}_1) = \frac{e^2}{4\pi\epsilon_0} \int \frac{1}{|\vec{r}_1 - \vec{r}_2|} |\psi_0(\vec{r}_2)|^2 d\vec{r}_2$$

Substitution leads to

$$E^{(1)} = \frac{e^2}{4\pi\epsilon_0} \int \psi_0^*(\vec{r}_1) \psi_0^*(\vec{r}_2) \frac{1}{|\vec{r}_1 - \vec{r}_2|} \psi_0(\vec{r}_1) \psi_0(\vec{r}_2) d\vec{r}_1 d\vec{r}_2 = J_{00}$$

$$\rightarrow \epsilon_0 = E_0 + J_{00}$$

Orbital Energies

- Ground-state energy $E = 2E_0 + J_{00}$
- Orbital energy $\varepsilon_0 = E_0 + J_{00}$

$$E \neq 2\varepsilon_0$$

(total energy of He $\neq$ sum of orbital energies)

Koopman's Approximation

- Ground-state energy $E = 2E_0 + J_{00}$
- Orbital energy $\varepsilon_0 = E_0 + J_{00}$

$$E = E_0 + \varepsilon_0 \rightarrow \varepsilon_0 = E - E_0$$

- E : energy of He atom
- E_0 : energy of He^+ ion (hydrogen-like atom with $Z = 2$)
- $E_0 - E$: ionization energy (IE) of He atom

$$\text{IE} \approx -\varepsilon_0$$

Case of Three Electrons (Li)

$$\phi_{ijk}(1, 2, 3) = \psi_i(\vec{r}_1)\psi_j(\vec{r}_2)\psi_k(\vec{r}_3)|\sigma_1\rangle_1|\sigma_2\rangle_2|\sigma_3\rangle_3$$

- Separation of spatial and spin variables: one cannot find a symmetric/antisymmetric spin wavefunction
- Need a systematic way to find a total antisymmetric wavefunction

Slater Determinant

- **Two-electron system:** for two quantum states i and j ,

$$\phi_{ij}(1, 2) = \frac{1}{\sqrt{2}} \begin{vmatrix} u_i(1) & u_j(1) \\ u_i(2) & u_j(2) \end{vmatrix}$$

- **N -electron system:** for N quantum states $a, b, \dots, n$,

$$\phi_{ab\dots n}(1, 2, \dots, N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} u_a(1) & u_b(1) & \cdots & u_n(1) \\ u_a(2) & u_b(2) & \cdots & u_n(2) \\ \vdots & \vdots & \ddots & \vdots \\ u_a(N) & u_b(N) & \cdots & u_n(N) \end{vmatrix}$$

Determinant of a Square Matrix

$$\det \mathbf{A} = \begin{vmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{vmatrix}$$

- 2 by 2 determinant

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

Expanding the Determinant

1. Pick a row or a column (with many zeros)
2. Calculate **each element** $\times$ **sign** $\times$ **minor**
 - **Minor**: determinant after deleting the row and the column containing the element
 - **Sign**: counting the number of steps of one row or one column required to get from the upper left element
→ even number: (+), odd number: (-)
3. Sum them up across the row or the column
4. Repeat this up to a sum of 2 by 2 determinants



Example

Expand the 3 by 3 determinant of a general matrix $\mathbf{A}$.

$$\det \mathbf{A} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Pick the first row:

$$\det \mathbf{A} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Now we can apply the definition of the 2 by 2 determinant:

$$\det \mathbf{A} = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

Property 1 of Determinant

$$\phi_{ab\dots n}(1, 2, \dots, N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} u_a(1) & u_b(1) & \cdots & u_n(1) \\ u_a(2) & u_b(2) & \cdots & u_n(2) \\ \vdots & \vdots & \ddots & \vdots \\ u_a(N) & u_b(N) & \cdots & u_n(N) \end{vmatrix}$$

- If two rows or two columns are interchanged, the determinant = $(-1) \times$ (original determinant).

→ Antisymmetric property

Property 2 of Determinant

$$\phi_{ab\dots n}(1, 2, \dots, N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} u_a(1) & u_b(1) & \cdots & u_n(1) \\ u_a(2) & u_b(2) & \cdots & u_n(2) \\ \vdots & \vdots & \ddots & \vdots \\ u_a(N) & u_b(N) & \cdots & u_n(N) \end{vmatrix}$$

- If two rows or two columns are identical, the determinant has value zero.

→ Pauli exclusion principle

Lithium Again

- One-electron states: $\psi_0|\alpha\rangle$, $\psi_0|\beta\rangle$, $\psi_1|\alpha\rangle$
- Slater determinant

$$\phi(1, 2, 3) = \frac{1}{\sqrt{3!}} \begin{vmatrix} \psi_0(\vec{r}_1)|\alpha\rangle_1 & \psi_0(\vec{r}_1)|\beta\rangle_1 & \psi_1(\vec{r}_1)|\alpha\rangle_1 \\ \psi_0(\vec{r}_2)|\alpha\rangle_2 & \psi_0(\vec{r}_2)|\beta\rangle_2 & \psi_1(\vec{r}_2)|\alpha\rangle_2 \\ \psi_0(\vec{r}_3)|\alpha\rangle_3 & \psi_0(\vec{r}_3)|\beta\rangle_3 & \psi_1(\vec{r}_3)|\alpha\rangle_3 \end{vmatrix}$$

→ always antisymmetric!

Shorthand Notations

- α and β by unbarred and barred spatial wavefunctions

$$\begin{aligned}
 \phi(1, 2, 3) &= \frac{1}{\sqrt{3!}} \begin{vmatrix} \psi_0(\vec{r}_1)|\alpha\rangle_1 & \psi_0(\vec{r}_1)|\beta\rangle_1 & \psi_1(\vec{r}_1)|\alpha\rangle_1 \\ \psi_0(\vec{r}_2)|\alpha\rangle_2 & \psi_0(\vec{r}_2)|\beta\rangle_2 & \psi_1(\vec{r}_2)|\alpha\rangle_2 \\ \psi_0(\vec{r}_3)|\alpha\rangle_3 & \psi_0(\vec{r}_3)|\beta\rangle_3 & \psi_1(\vec{r}_3)|\alpha\rangle_3 \end{vmatrix} \\
 &= \frac{1}{\sqrt{3!}} \begin{vmatrix} \psi_0(1) & \overline{\psi_0(1)} & \psi_1(1) \\ \psi_0(2) & \overline{\psi_0(2)} & \psi_1(2) \\ \psi_0(3) & \overline{\psi_0(3)} & \psi_1(3) \end{vmatrix} \\
 &= |\psi_0 \quad \overline{\psi_0} \quad \psi_1|
 \end{aligned}$$

Hartree-Fock Method

- The self-consistent field (SCF) method
- The atomic wavefunction = Slater determinant
- Electrons interact only through an average potential
→ They are “uncorrelated”
- (Electron) correlation energy E_{corr}

$$E_{\text{corr}} = E_{\text{exact}} - E_{\text{HF}}$$

N -Electron Hamiltonian

For an N -electron atom,

$$\hat{H} = \sum_{j=1}^N \left(-\frac{\hbar^2}{2m_e} \nabla_j^2 - \frac{Ze^2}{4\pi\epsilon_0 r_j} \right) + \sum_{j=1}^N \sum_{i<j} \frac{e^2}{4\pi\epsilon_0 |\vec{r}_i - \vec{r}_j|}$$

or

$$\hat{H} = \sum_{j=1}^N \hat{h}_j + \sum_{j=1}^N \sum_{i<j} \frac{e^2}{4\pi\epsilon_0 |\vec{r}_i - \vec{r}_j|}$$

Ground-State Wavefunction

With $N = 2M$, the ground-state wavefunction is

$$\phi(1, 2, \dots, 2M) = |\psi_0 \quad \overline{\psi_0} \quad \psi_1 \quad \overline{\psi_1} \quad \cdots \quad \psi_{M-1} \quad \overline{\psi_{M-1}}|$$

$$\hat{H} = \underbrace{\sum_{j=1}^{2M} \hat{h}_j}_{\hat{H}_0} + \underbrace{\sum_{j=1}^{2M} \sum_{i < j} \frac{e^2}{4\pi\epsilon_0 |\vec{r}_i - \vec{r}_j|}}_{\hat{H}'}$$

Ground-State Energy

- Zeroth-order term

$$\hat{H}_0 \phi(1, 2, \dots, 2M) = \left(\sum_{j=1}^M 2E_j \right) \phi(1, 2, \dots, 2M)$$

- First-order term

$$E^{(1)} = \int \phi^*(1, 2, \dots, 2M) \hat{H}' \phi(1, 2, \dots, 2M) d\vec{r}_1 d\vec{r}_2 \cdots d\vec{r}_{2M}$$

First-Order Term

$$E^{(1)} = \int \phi^*(1, 2, \dots, 2M) \hat{H}' \phi(1, 2, \dots, 2M) d\vec{r}_1 d\vec{r}_2 \cdots d\vec{r}_{2M}$$

$$J_{ij} = \int \psi_i^*(\vec{p}) \psi_j^*(\vec{q}) \frac{e^2}{4\pi\epsilon_0 |\vec{p} - \vec{q}|} \psi_i(\vec{p}) \psi_j(\vec{q}) d\vec{p} d\vec{q}$$

$$K_{ij} = \int \psi_i^*(\vec{p}) \psi_j^*(\vec{q}) \frac{e^2}{4\pi\epsilon_0 |\vec{p} - \vec{q}|} \psi_i(\vec{q}) \psi_j(\vec{p}) d\vec{p} d\vec{q}$$

$$E^{(1)} = \sum_{i=1}^M \sum_{j=1}^M (2J_{ij} - K_{ij})$$

Ground-State Energy

$$E = \sum_{j=1}^M 2E_j + \sum_{i=1}^M \sum_{j=1}^M (2J_{ij} - K_{ij})$$

Similarly, the orbital energy can be shown to be

$$\varepsilon_i = E_i + \sum_{j=1}^M (2J_{ij} - K_{ij})$$

$$\therefore E = \sum_{i=1}^M (E_i + \varepsilon_i)$$

RS 9.5
M 9.7

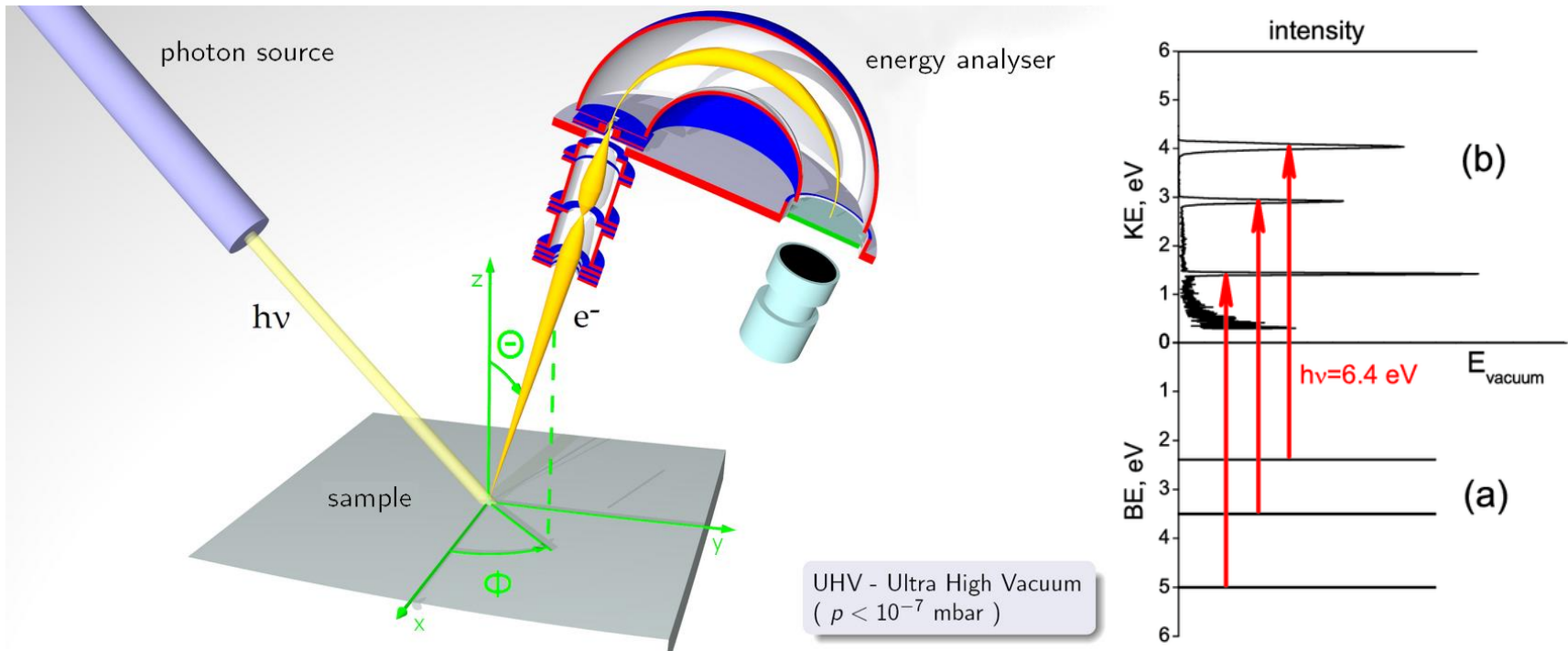
Koopman's Approximation

$$IE \approx -\varepsilon_i$$

- Good approximation for closed-shell systems
- Ionization energy gives information about orbital energies

Not in text

Photoelectron Spectroscopy



Ionization Energies

Electron removed	Resulting orbital occupancy	Ionization energies / MJ·mol ⁻¹		
		Koopman's approximation	HF calculation	Experimental
Neon				
1s	1s ¹ 2s ² 2p ⁶	86.0	83.80	83.96
2s	1s ² 2s ¹ 2p ⁶	5.07	4.76	4.68
2p	1s ² 2s ² 2p ⁵	2.23	1.92	2.08
Argon				
1s	1s ¹ 2s ² 2p ⁶ 3s ² 3p ⁶	311.4	308.25	309.32
2s	1s ² 2s ¹ 2p ⁶ 3s ² 3p ⁶	32.35	31.33	
2p	1s ² 2s ² 2p ⁵ 3s ² 3p ⁶	25.13	24.01	23.97
3s	1s ² 2s ² 2p ⁶ 3s ¹ 3p ⁶	3.35	3.20	2.82
3p	1s ² 2s ² 2p ⁶ 3s ² 3p ⁵	1.55	1.43	1.52